

### Hierarchy of Evidence:

1. Systematic Review: secondary study, filtered info
2. Meta Analysis: secondary study, filtered info
3. Randomized Control Trial (RCT): primary study, unfiltered info
4. Cohort Study: primary study, unfiltered info
5. Case Control Study: primary study, unfiltered info
6. Case Report: primary study, unfiltered info
7. Expert opinion/background info: primary study, unfiltered info

\*quality improves higher up on the hierarchy, risk of bias increases as you move further down. Support for evidence improves higher up in the hierarchy\*

1. Empirical evidence: info gathered via direct observation and experimentation, relies on sensory data (case studies)
2. Bias: assumption which affects how you view something unconsciously/consciously, positively/negatively
3. Confounding: a third variable influences both the independent and dependent variable, making it unclear if the relationship between them is real/accurate
4. Primary studies: collecting and analyzing first hand data to a specific research question
5. Secondary studies: analyzes existing research/data instead of collecting new data
6. Association: idea that factors/variables are statistically related
7. Reliability: an individual can trust a measurement has been tested repeatedly, producing same, consistent outcome
8. Reproducibility: getting same results on same data with exact same methods used in context of health research
9. Correlation: mutual relationship between two things, how they move together
10. Causation: one thing directly influencing/changing something else (ex. Smoking and lung health)
11. Validity: how accurately a method measures what it is intended to measure
  - Internal validity: how well a study's design prevents alternative explanations for results
  - External validity: extent to which findings apply to other people, settings, time

### Evidence Based Practice:

- Decision-making approach that combines **best available research evidence** with professional expertise and stakeholder values
- Health disciplines should be based on scientific research
- Minimize outdated/unsound/unethical practices in favor of those more effective

### Best available research evidence:

1. Meta analysis: type of systematic review that combines (pools) results using accepted statistical methodology as if they were from 1 large study
2. Systematic review: focus on specific clinical question, includes extensive/explicit review of literature to identify studies with sound methodology
  - Data extracted from study are combined, compared, assessed
  - Conclusions are made based on results and/or presence or absence of supporting evidence

#### Retracted 1998 Wakefield 'MMR-Autism' Paper:

- Claim: In 1998, Andrew Wakefield and 12 colleagues published study in The Lancet suggesting link between the Measles, Mumps, and Rubella (MMR) vaccine and a syndrome they called 'autistic enterocolitis'
- Impact: significant decline in vaccine uptake, rise in preventable diseases
- Scientific and Ethical Failures: sample size of only 12 children, journalistic misrepresentation, conflict of interest, patient harm
- Outcome: The Lancet issued a retraction in 2010. Subsequent studies confirmed there is no link between vaccines and autism

#### Damage done by 1998 Wakefield Zombie paper:

- Hundreds of citations of the retracted Wakefield paper
- Surprisingly, referencing increased post-retraction
- 2011-2017: 337 academic citations of the Wakefield study. 62% failed to note that it was retracted, 50% were negative in tone
- Paper continually reanimated by anti-vaccine activists

#### Limitations of Retracted Studies/Retraction:

- Only 5.4% of post-retraction references acknowledge retraction
- Of 20,000 retracted articles, 84% received at least 1 post-retraction citation

#### What factors are associated with COVID-19 vaccine hesitancy?

- Conspiracy theories: sentiment of distrust in govs, healthcare institutions. Once limited to fringe audiences (edge of society), have now become common in mass media, the internet, social media
- Health Belief Model: People weigh severity of the health threat they confront and the perceived benefits/harms of taking a specific action
  - Individual assessment of one's own risk can be influenced by the information received and the social context in which the individual lives and interacts
  - Many people seek information that supports their convictions (confirmation bias)
  - Misinformation is frequently shared in echo chambers of like-minded groups of individuals (social expectation)